

TrimNN: characterizing cellular community motifs for studying multicellular topological organization in complex tissues

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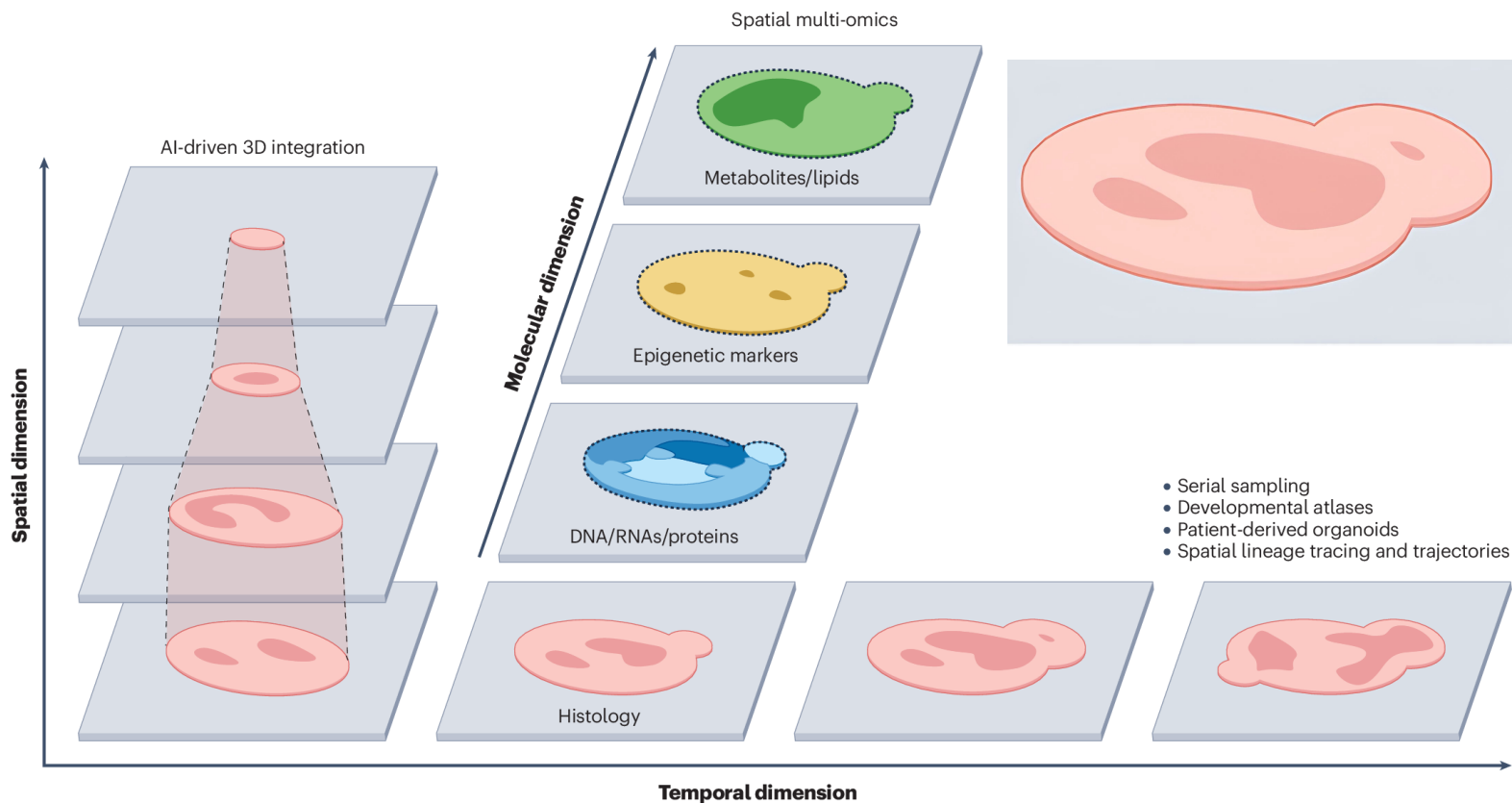
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07/07/2026



Multi-omics in spatial temporal to decode complex biological system



- **Spatial omics** technologies, encompassing both imaging and sequencing-based methodologies, has enabled a **comprehensive understanding of biological processes from a cellular ecosystem perspective.**

- **Biggest computational challenge: integration**

Lázár, E., Lundeberg, J. Spatial architecture of development and disease. *Nat Rev Genet* (2025)

Video by Grok ²

Method of the year in a row

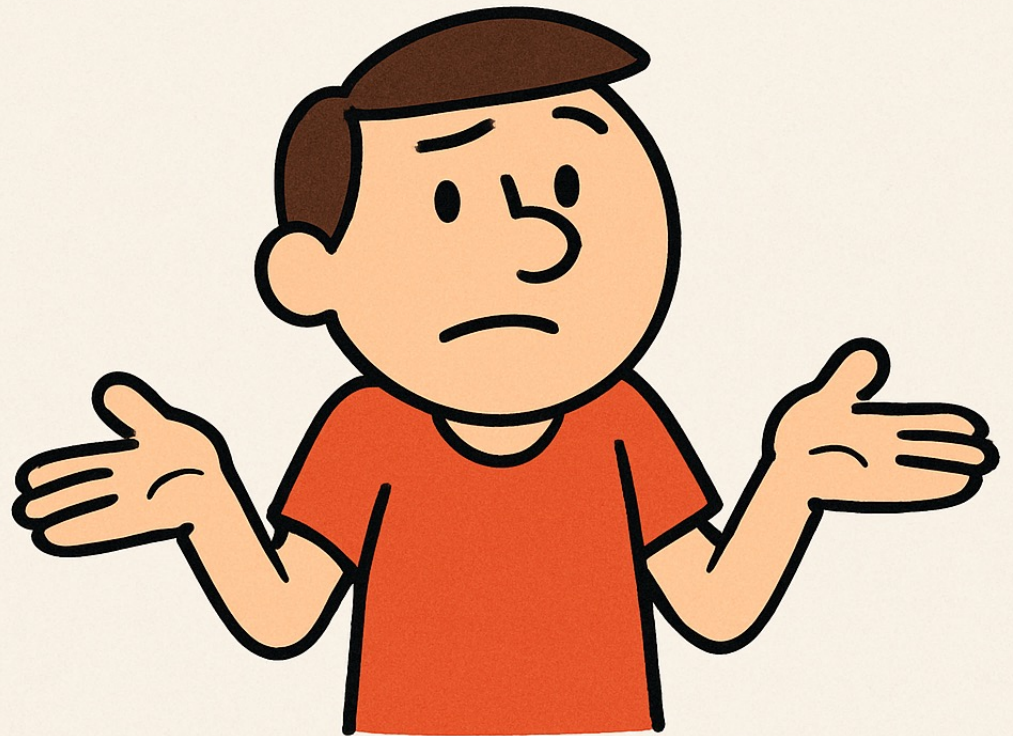


- 2013 **Single cell sequencing** (Method of the year by Nature Methods)
- 2019 **Single cell multimodal omics** (Method of the year by Nature Methods)
- 2020 **Spatially resolved transcriptomics** (Method of the year by Nature Methods)
- 2022 **Spatial Omics**: one of the top seven technologies to watch (Nature)
- 2024 **Spatial proteomics** (Method of the year by Nature Methods)

Question:

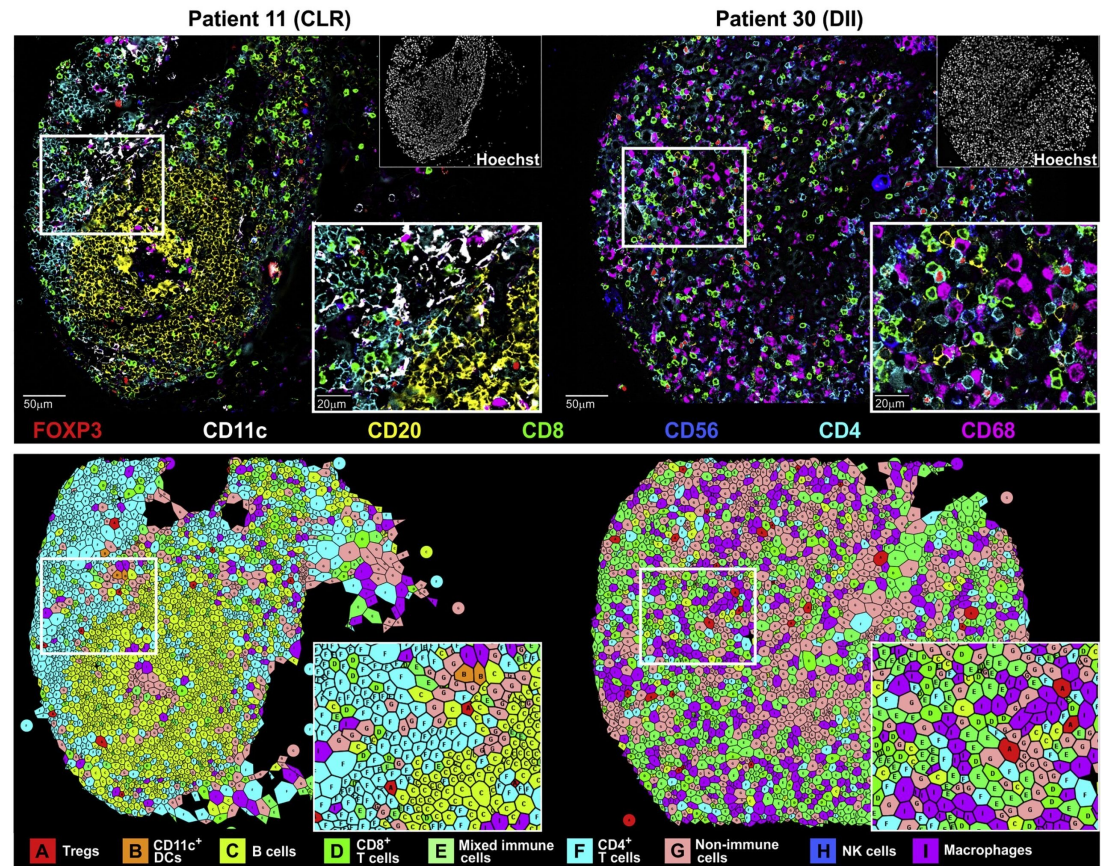
- How and why are cells organized differently to perform diverse functions?

SO WHAT?



Spatial niche (cellular neighborhood) as the building blocks of cell organization

- **Cell niches** are groups of cells that are spatially co-localized and share similar functional or phenotypic characteristics.
 - Based on cell type composition, gene expression profiles, and spatial proximity
- Insights:
 - Quantitative characterization of niche activity
 - Inference of cell-cell communication
 - Understanding of tissue architecture and organization
 - Identification of disease-relevant microenvironments



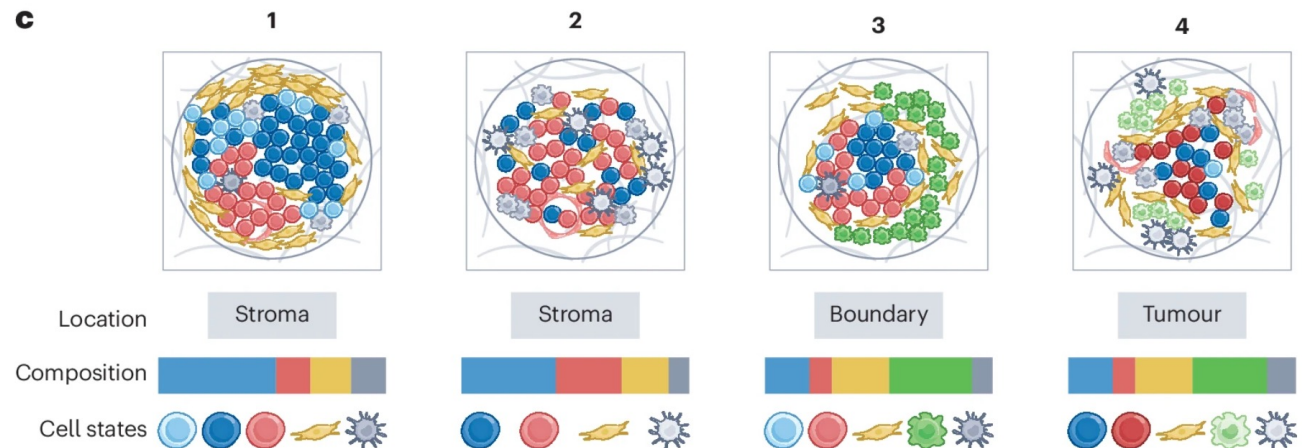
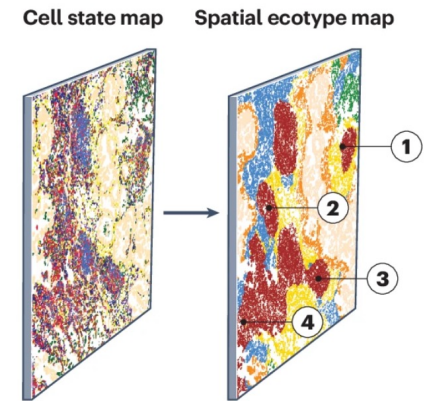
Schürch, et al. "Coordinated cellular neighborhoods orchestrate antitumoral immunity at the colorectal cancer invasive front." *Cell* 182.5 (2020): 1341-1359.

Limitations of top-down approaches

- Modeling cellular communities as a graph
- Clustering on learnt embedding
 - SpaceGM
 - CytoCommunity
 - BANKSY
 - ...
- Pros: Fast
- Cons:
 - Expressibility
 - Clustering
 - Interpretability
 - Generalizability

Can we have an alternative bottom-up view?

CNs from four different tertiary lymphoid structures (TLS) located within the tumor or surrounding stroma

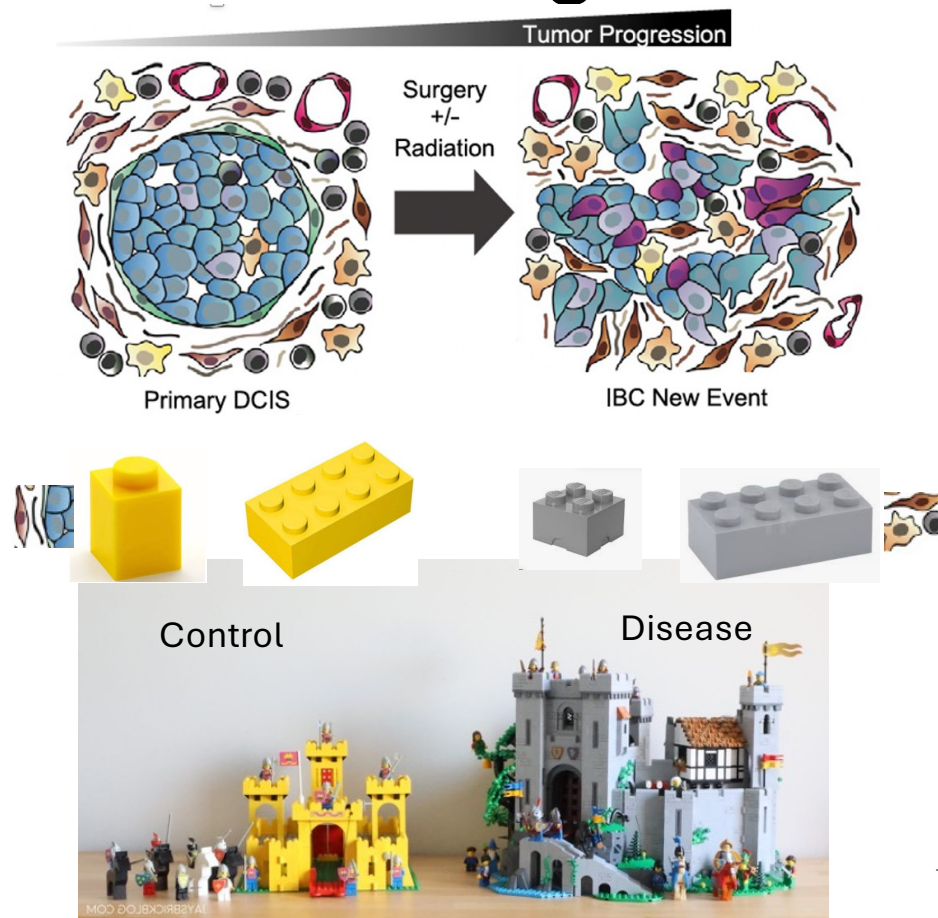


1 and 2 share **similar composition of major cell types**, but their detailed structural organization and cell states are different

Conceptional: Quantify cell organization as building blocks

- **Central hypothesis:** There are conservative patterns as building blocks in cell organization that determine cellular function.
 - Topology->gene activity/cell-cell communications->cellular function
- Can we answer these questions?
 - Can we *quantify* the patterns of cell organization?
 - Can we *interpret* the relation between the patterns and biological contexts?
 - Can we *predict* the outcome of disturbing the patterns?
 - Can we *design* patterns for specific outcomes?

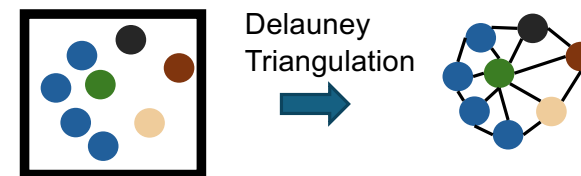
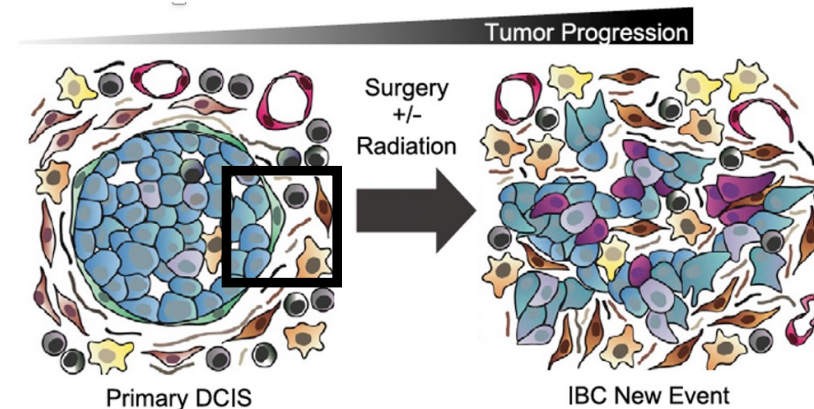
Risom et al., 2022, Cell 185, 299–310



Conceptual to formal:

Cellular Community motifs to quantify cell organization as building blocks

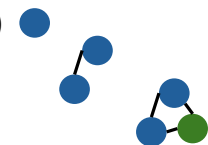
- **Hypothesis:** CC motifs can quantify cell organization in spatial omics data.
 - **Node** as cell, node label as cell type
 - **Edge** as cell relation, spatial or hypothetical relation, no direction
- CC motifs are **conservative** and can be **built up from scratch**.
- CC motifs are related to FTU but not identical.



Cells: **size-1** (Cell type composition)

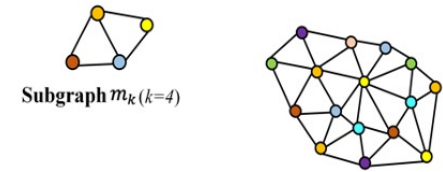
Edges: **size-2** (Cell-cell proximity)

Triangles: **size-3** (Proposed here)



Bottom-up challenges:

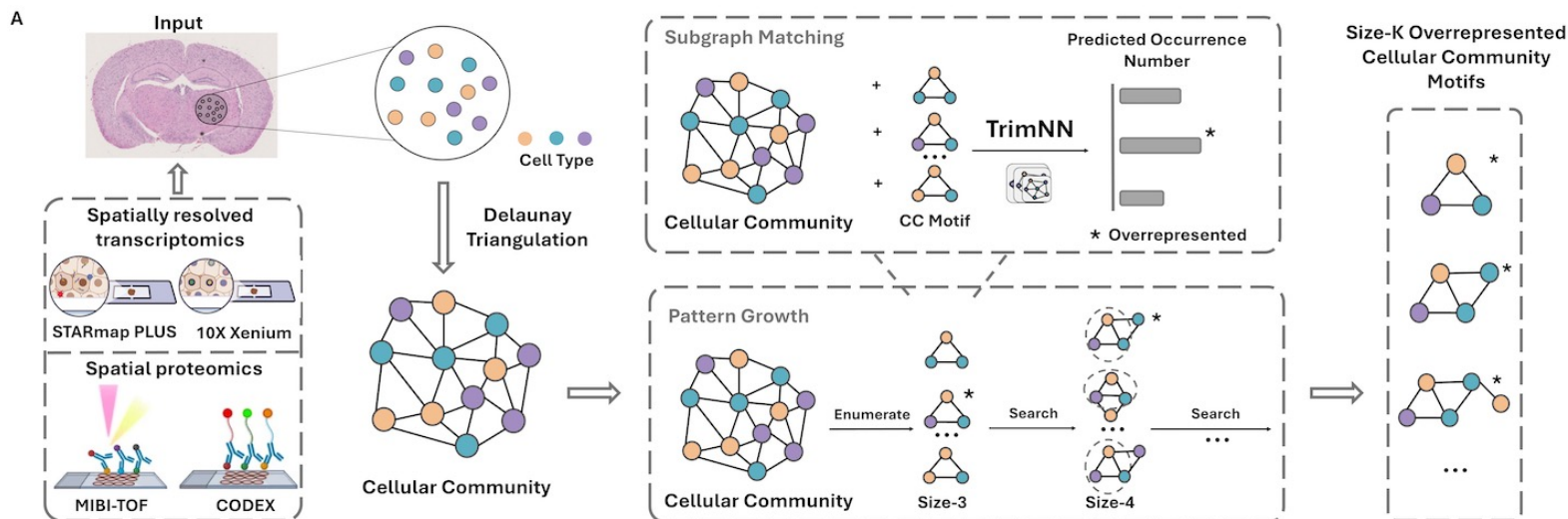
Finding most frequent size- k motifs



- **Goal:** Identify, among all possible graphs of k nodes, find the query graphs with the highest frequency in the target graph.
- Two sub-challenges:
 - **1) Subgraph matching:** given a specific subgraph m_k , counts the occurrence of isomorphic subgraphs within the target graph G . $F(G, m_k)$
 - **2) Pattern growth:** enumerates M_k and identifies the most overrepresented subgraphs based on their occurrences in the **subgraph matching**. $F(G, M_k)$
- **Infeasible** to calculate large-size subgraphs
 - Subgraph isomorphism is **NP-complete** + Node type combination
 - Computation time *grows exponentially* as the size of the subgraphs increases
 - Feasible motif size for traditional methods is relatively small (1 to 3)
 - Edge sampling (e.g., MFinder), node sampling (e.g., FANMOD), and global pruning (e.g., Ullmann and VF2)

Can ML help to identify large-size motifs?

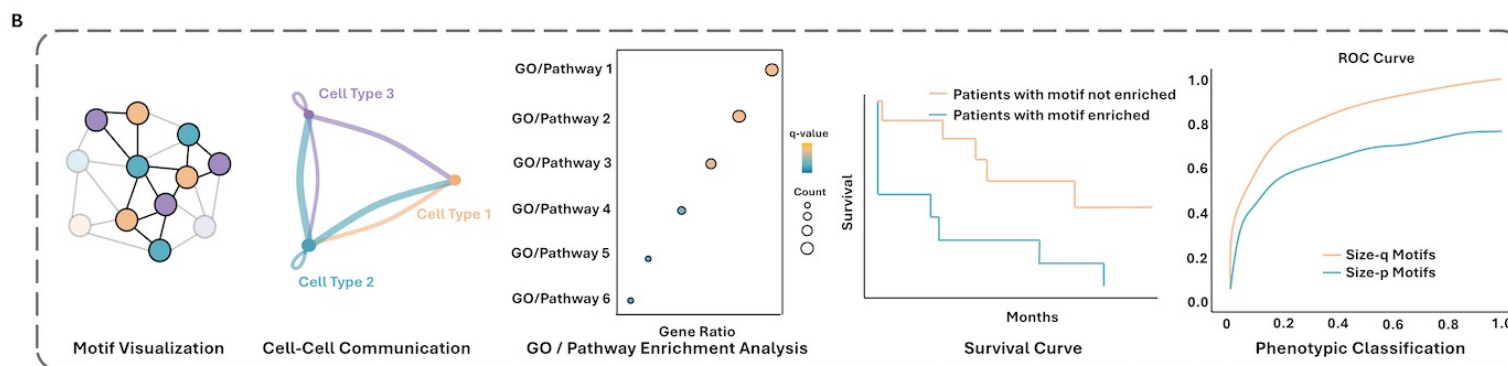
TrimNN: Characterizing cellular community motifs for studying multicellular topological organization in complex tissues



Yang Yu
U of Missouri

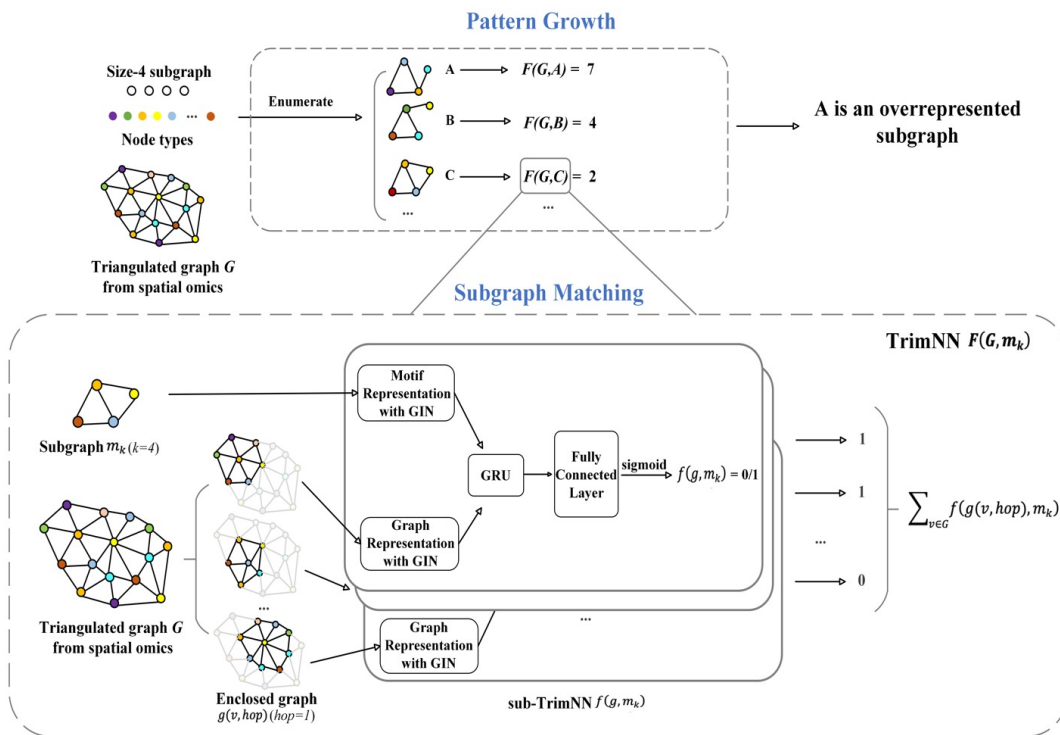


Shuang Wang
Indiana University



Yang Yu, Shuang Wang, Jinpu Li, Meichen Yu, Kyle McCrocklin, Jing-Qiong Kang, Anjun Ma, Qin Ma, Dong Xu, **Juexin Wang**, TrimNN: Characterizing cellular community motifs for studying multicellular topological organization in complex tissues, *Nat Commun* 16, 7737 (2025)

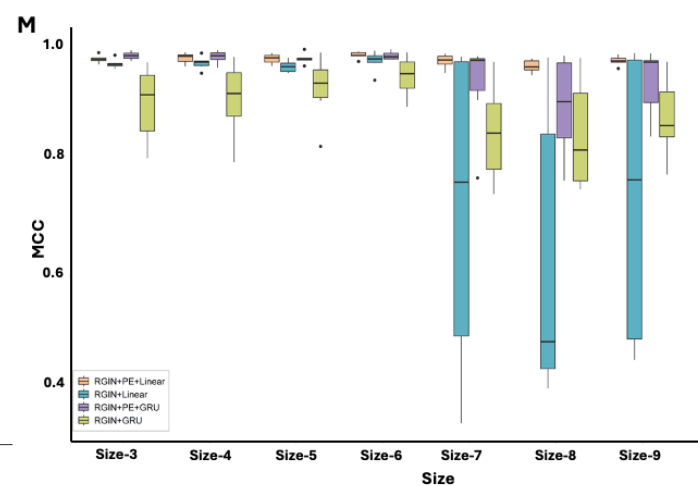
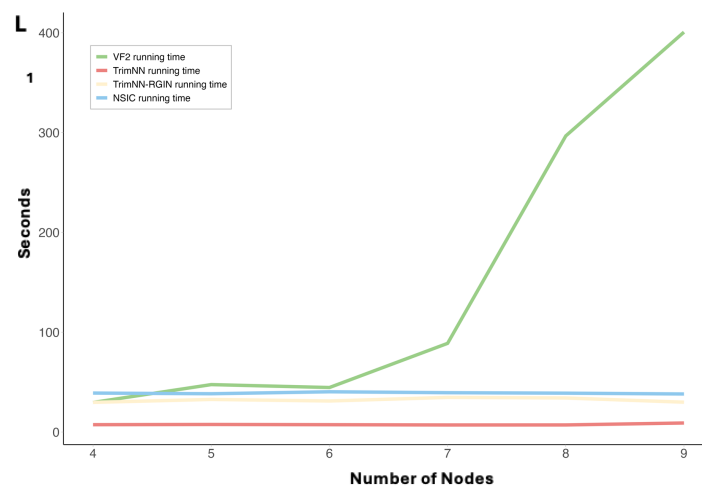
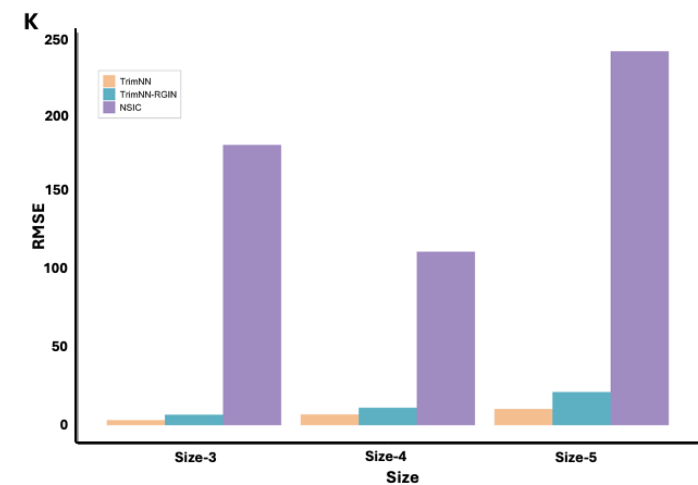
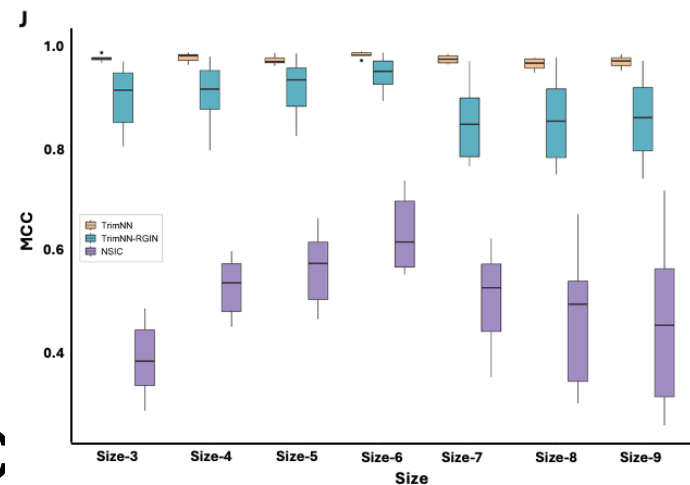
Triangulation cellular community Motif Neural Network (TrimNN)



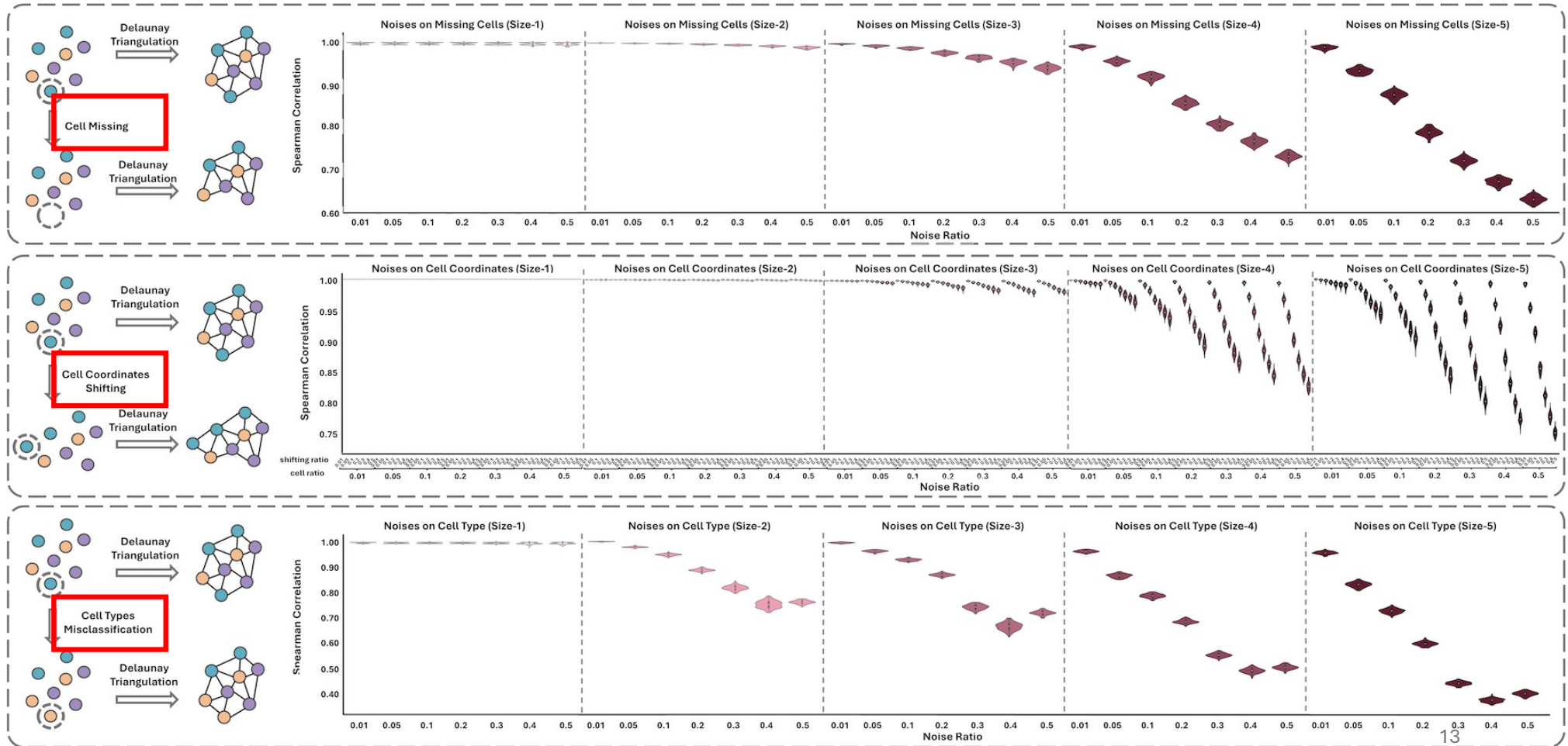
Yang Yu, Shuang Wang, Dong Xu, **Juexin Wang**, Exploring building blocks of cell organization by estimating network motifs using graph isomorphism network, **NeurIPS 2023 GenBio workshop**, Spotlight Oral

- Estimate the prevalence of CC motifs of any size in a spatial omics induced graph.
- **Biologically meaningful constrained problem**
 - Aims to address the **Subgraph Matching** problem in *triangulated graphs* derived from spatial omics.
- **Quasi-divide and conquer**
 - *Decomposes* the occurrence regression challenge into several binary classification problems modeled by the sub-TrimNN module.
- **Estimate relative abundance**
 - Aggregates the sub-TrimNN module's results and outputs the subgraphs' *relative abundance*
- **Supervised model**
 - Trained on representative pairs of the predefined subgraphs and the triangulated cell graphs.

TrimNN accurately
identified
overrepresented CC
motifs at spatial
omics level



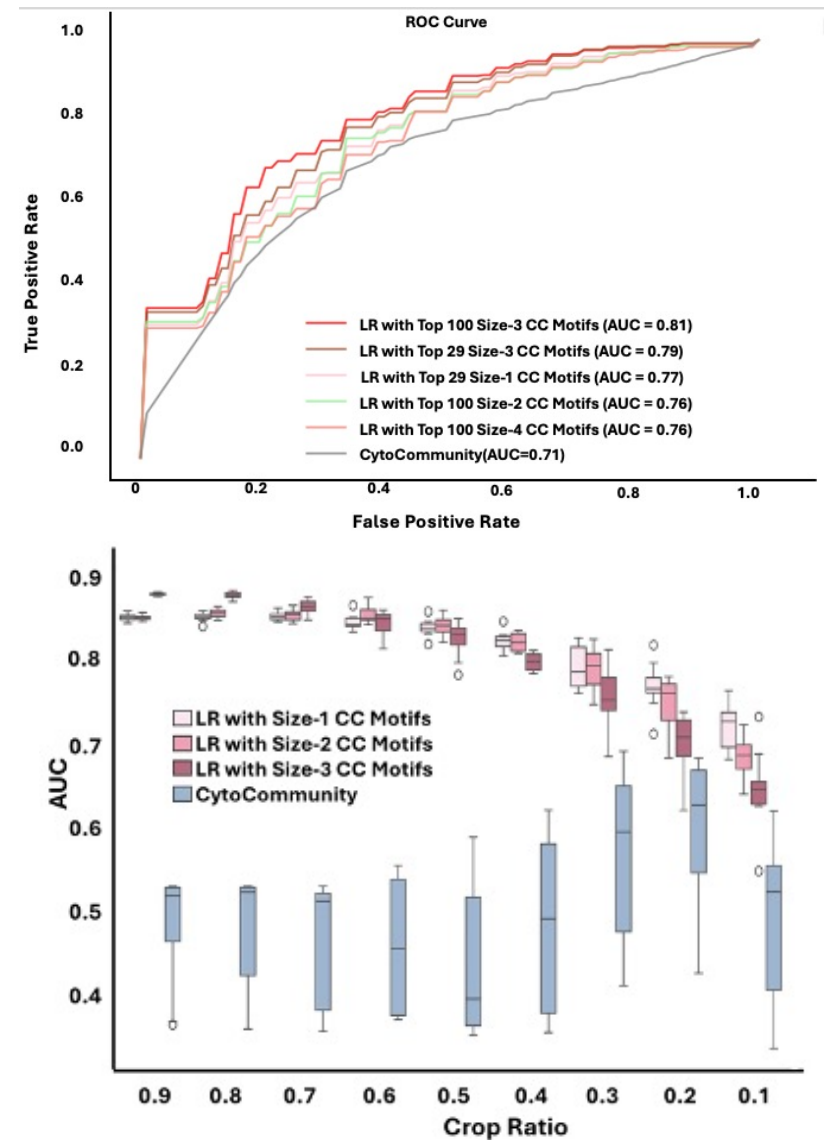
CC motifs are **robust** in Delaunay triangulated graph from spatial omics



CC motifs as features for better **generalizability** cross different samples

- Colorectal cancer using CODEX
 - 17 low-risk (Crohn's-like lymphoid reaction (CLR))
 - 18 high-risk (diffuse inflammatory infiltration (DII))
- Robust in new samples by random cropping
- Similar results in RF, SVM
- Fewer features
- Simpler model
- Interpretable model

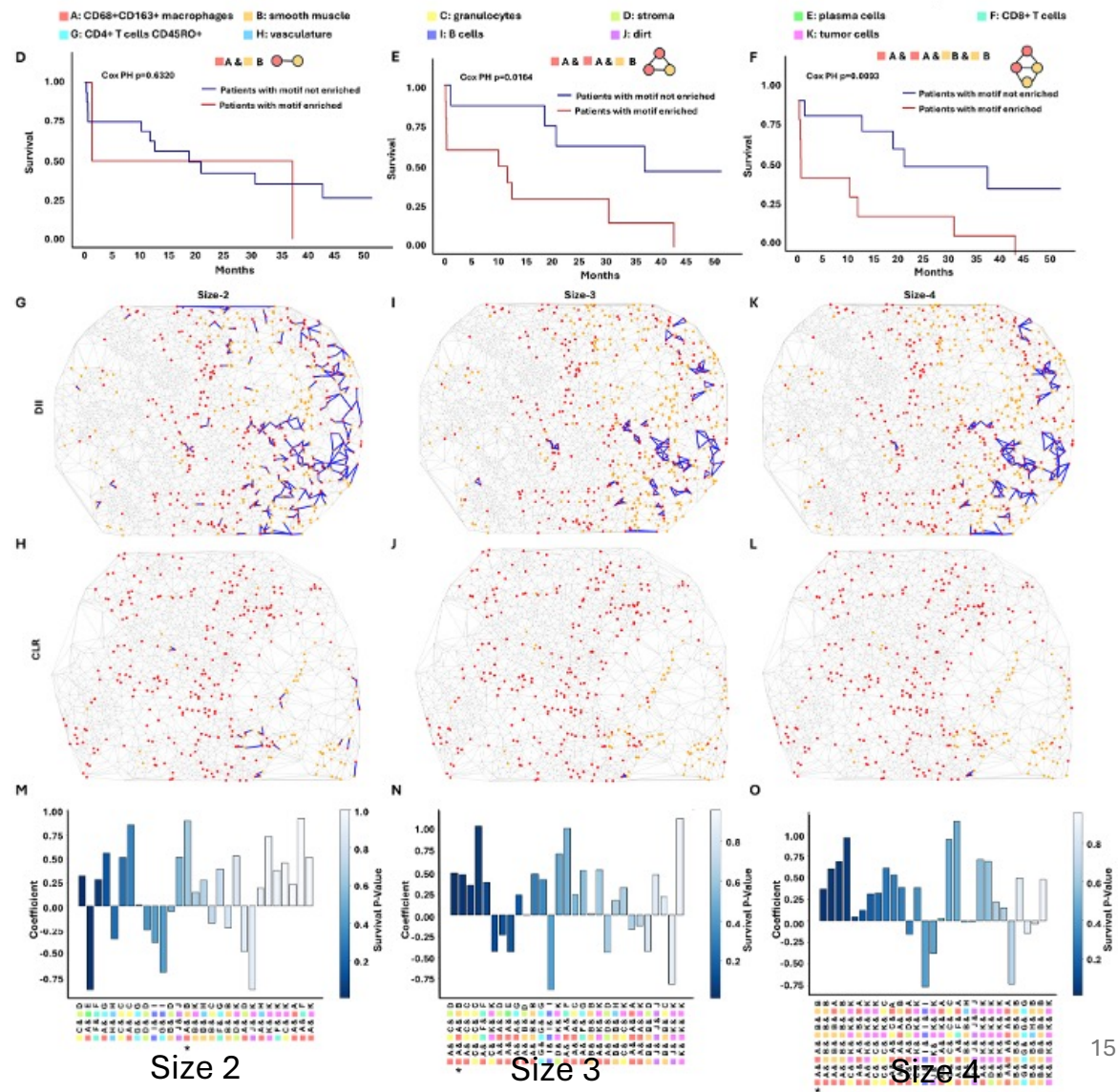
Schürch, Christian M., et al. "Coordinated cellular neighborhoods orchestrate antitumoral immunity at the colorectal cancer invasive front." *Cell* 182.5 (2020): 1341-1359.



- Enriched CC motifs can differentiate survival in various sizes:

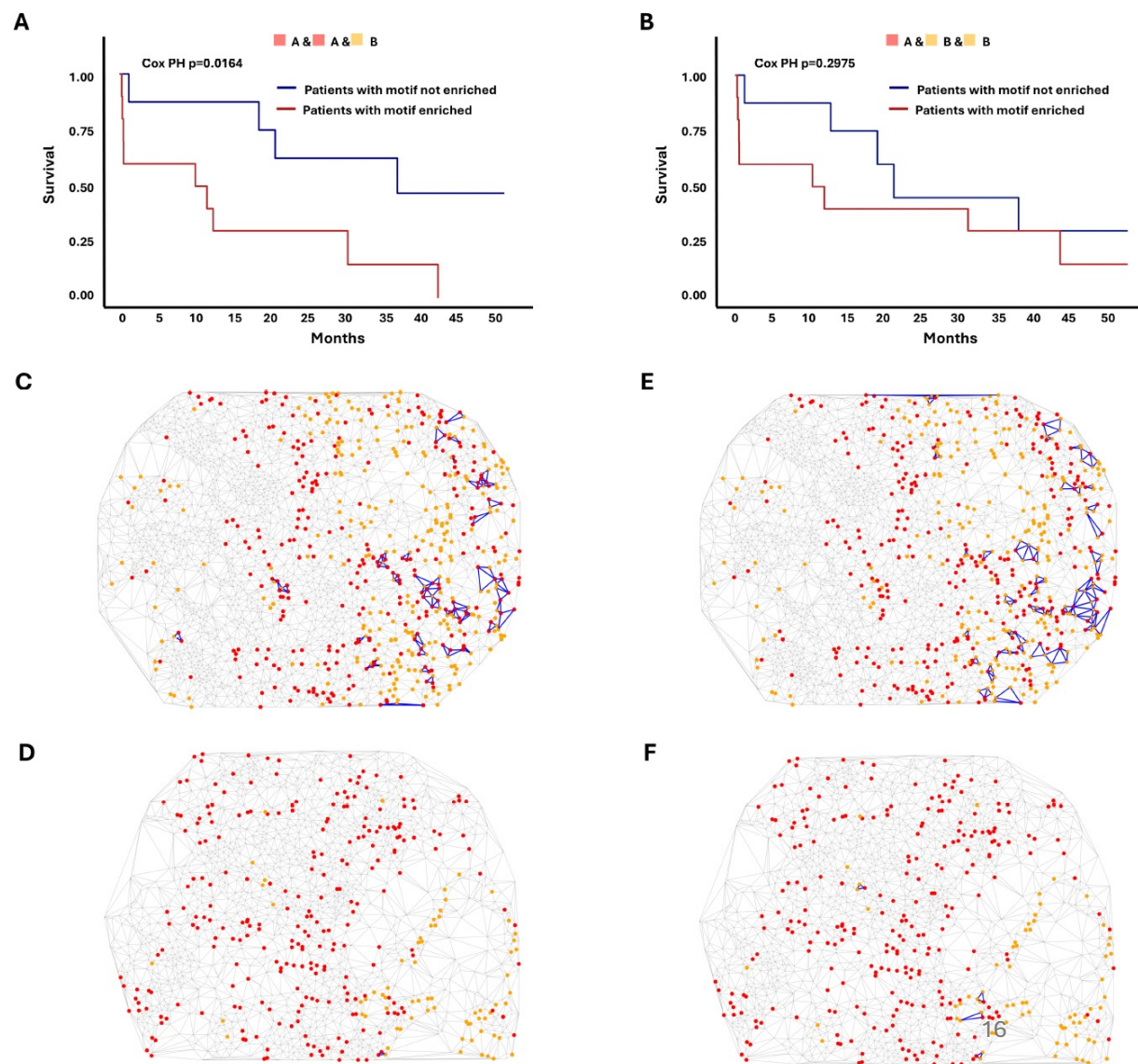
- Size2: Marcophage + Muscle
- Size 3: Marcophage + Macrophage+Muscle
- Size 4: Marcophage+ Marcophage+Muscle +Muscle

- Interpretability as coefficients from LR



Spatial topology plays a crucial role in linking phenotypes and survival (Expressivity)

- CD68+CD163+macrophages ('A')
- smooth muscle ('B')
- '**A & A & B**' had occurred 4,176 and 1,548 times among DII and CLR patients with a ratio of 2.70, clearly differentiate survival (Cox PH=0.0164)
- '**A & B & B**' had occurred 4,602 and 2,255 times among DII and CLR patients with a lower ratio of 2.04, and it cannot differentiate survival well (Cox PH = 0.2975)

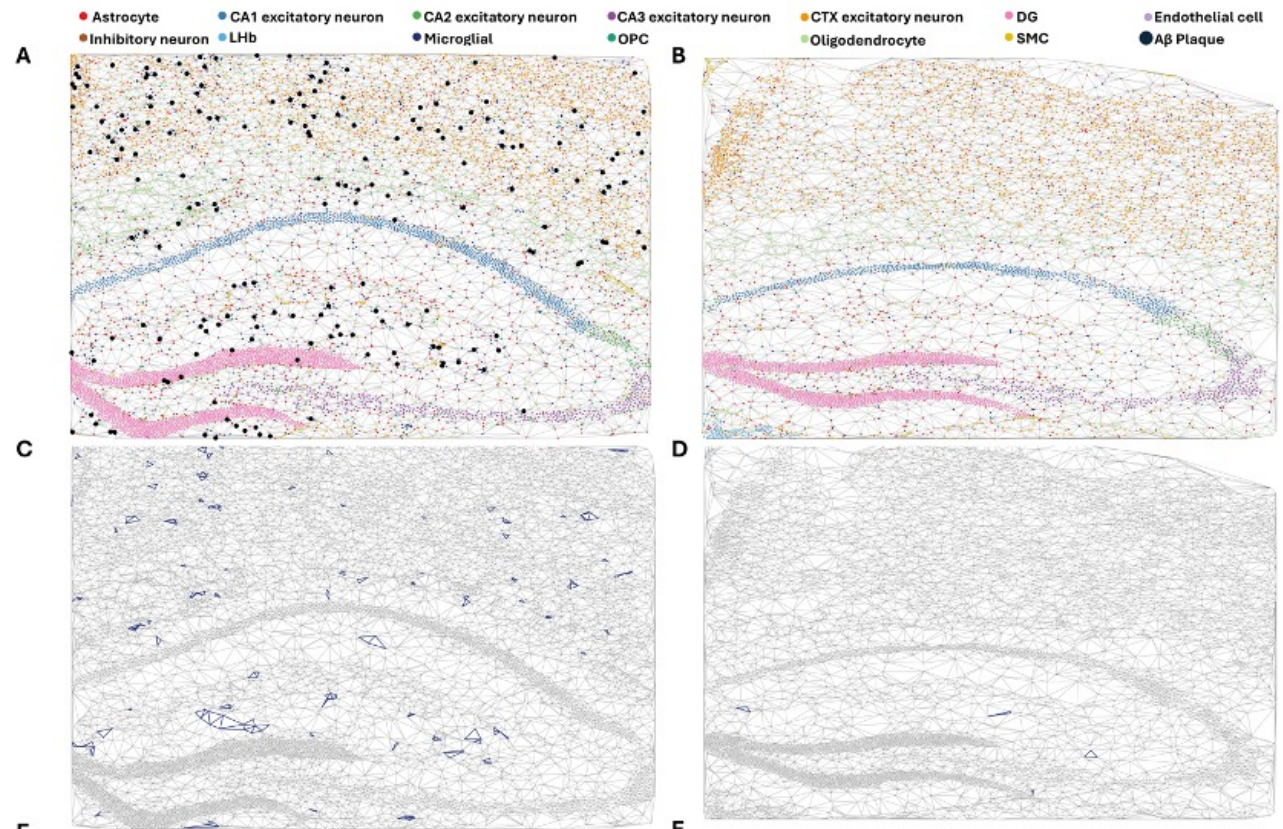


Diverse CC motifs in Alzheimer's disease

- Topological combinations of microglia and cortex excitatory neurons, particularly their relationship with amyloid-b on the cellular level, are still unknown in AD.

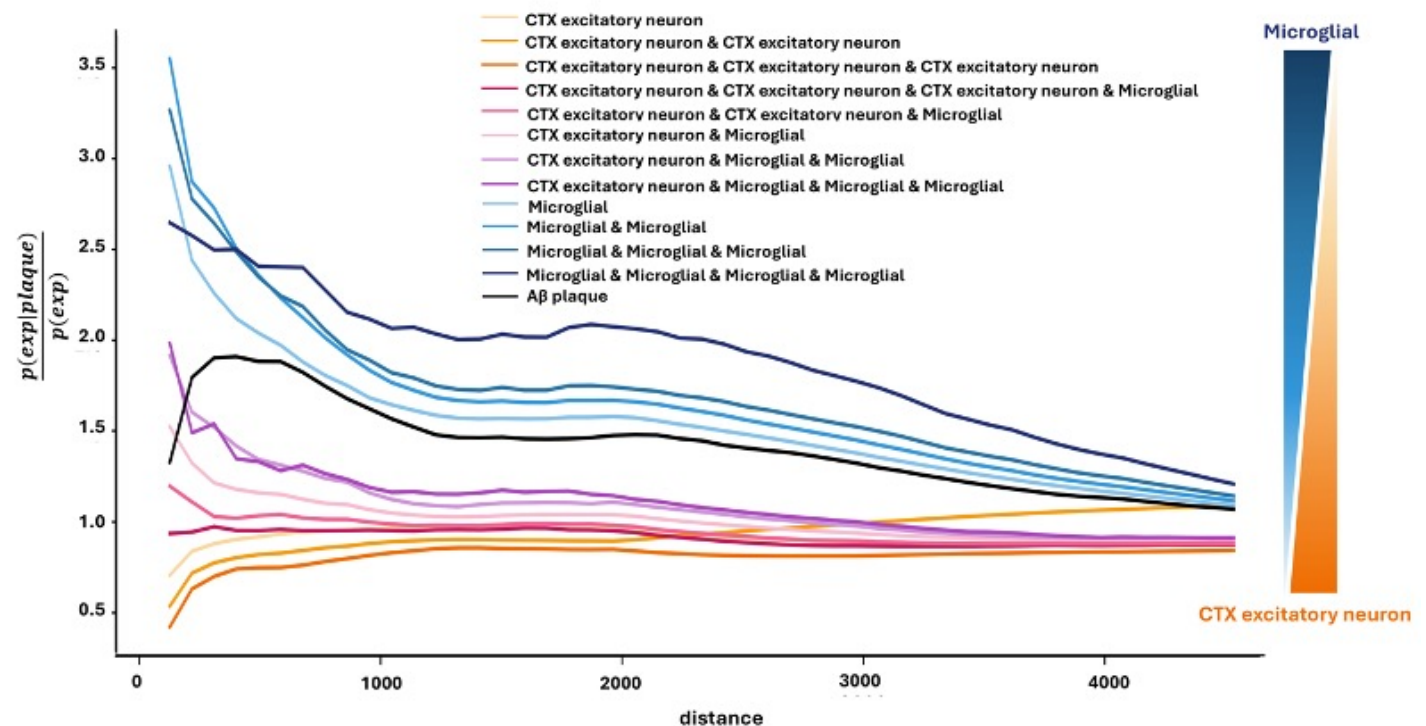
Spatial Transcriptomics: STARmap PLUS

Zeng, Hu, et al. "Integrative in situ mapping of single-cell transcriptional states and tissue histopathology in a mouse model of Alzheimer's disease." *Nature Neuroscience* 26.3 (2023): 430-446.



Two types of CC motifs in different mechanisms

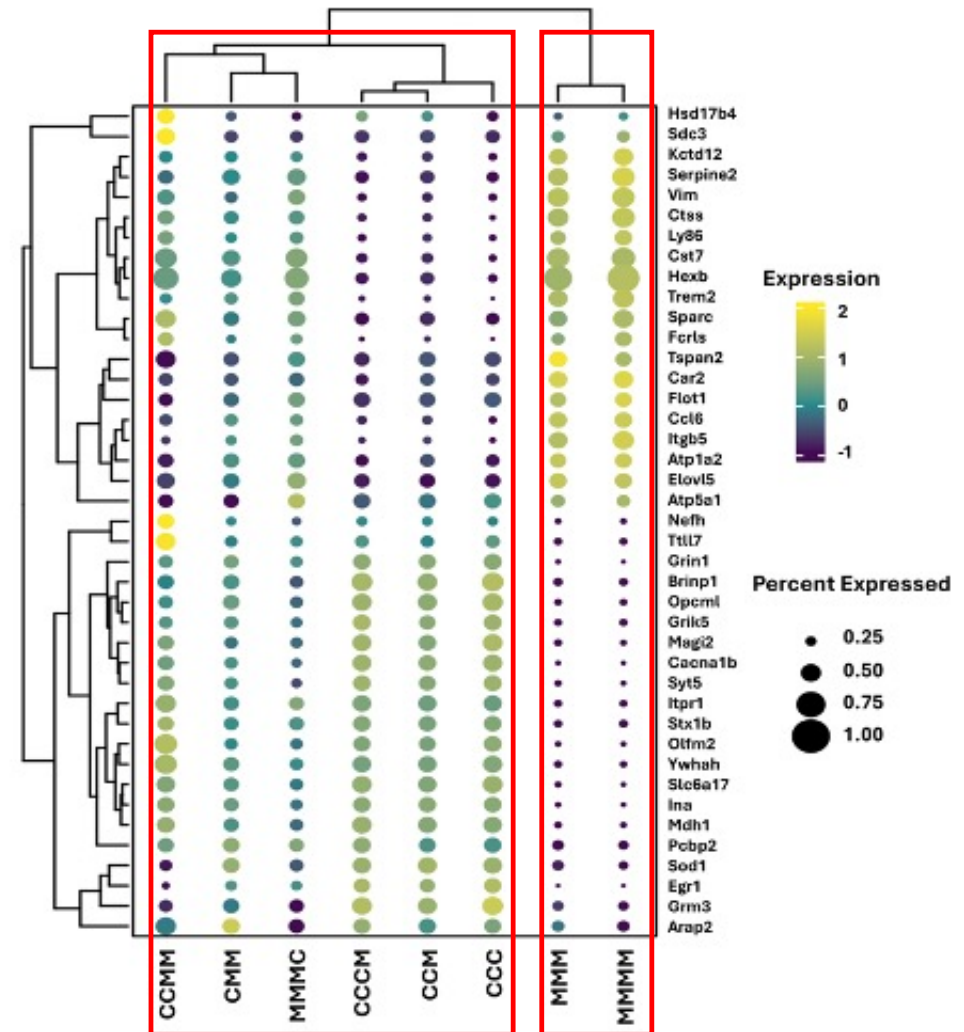
- Spatial co-occurrence analysis on **Spectra** of CC motifs composed of microglia and cortex excitatory neurons in multiple sizes
- By Squidpy
- Microglia activate inflammation in the vicinity of amyloid- β deposits, which are directly toxic to the adjacent neurons



- Microglia and Cortex excitatory neurons differ in cell-cell communications and marker gene analysis

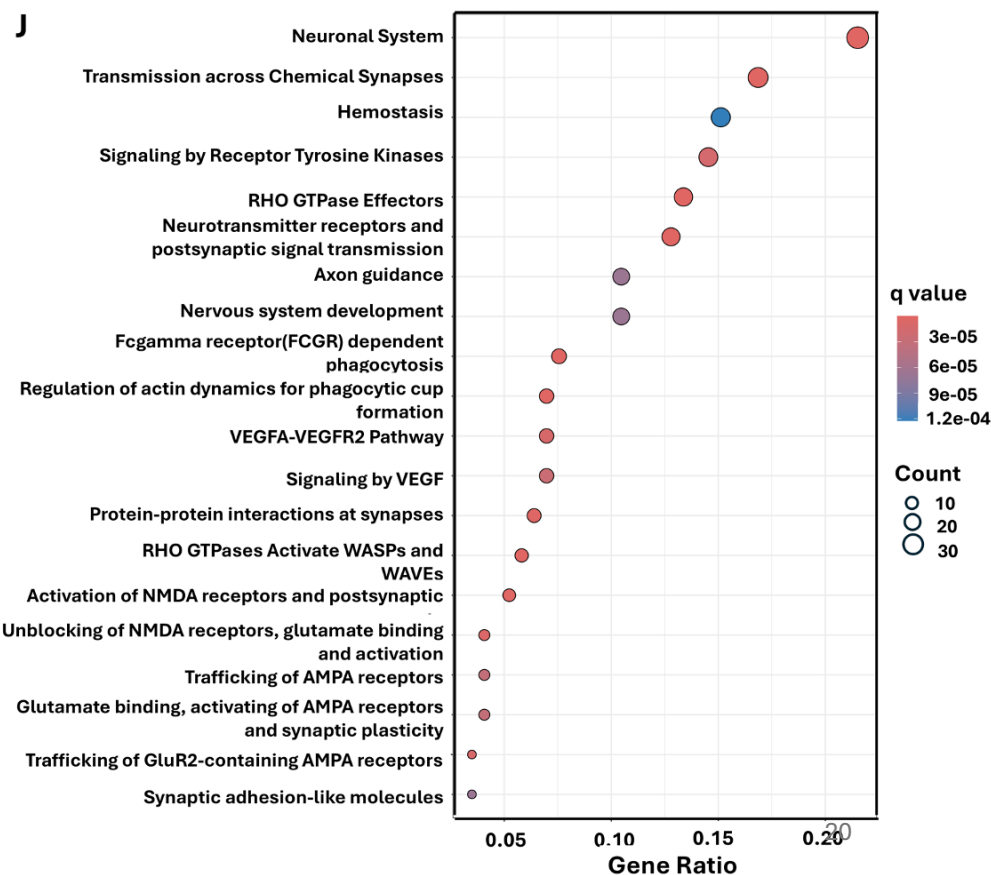
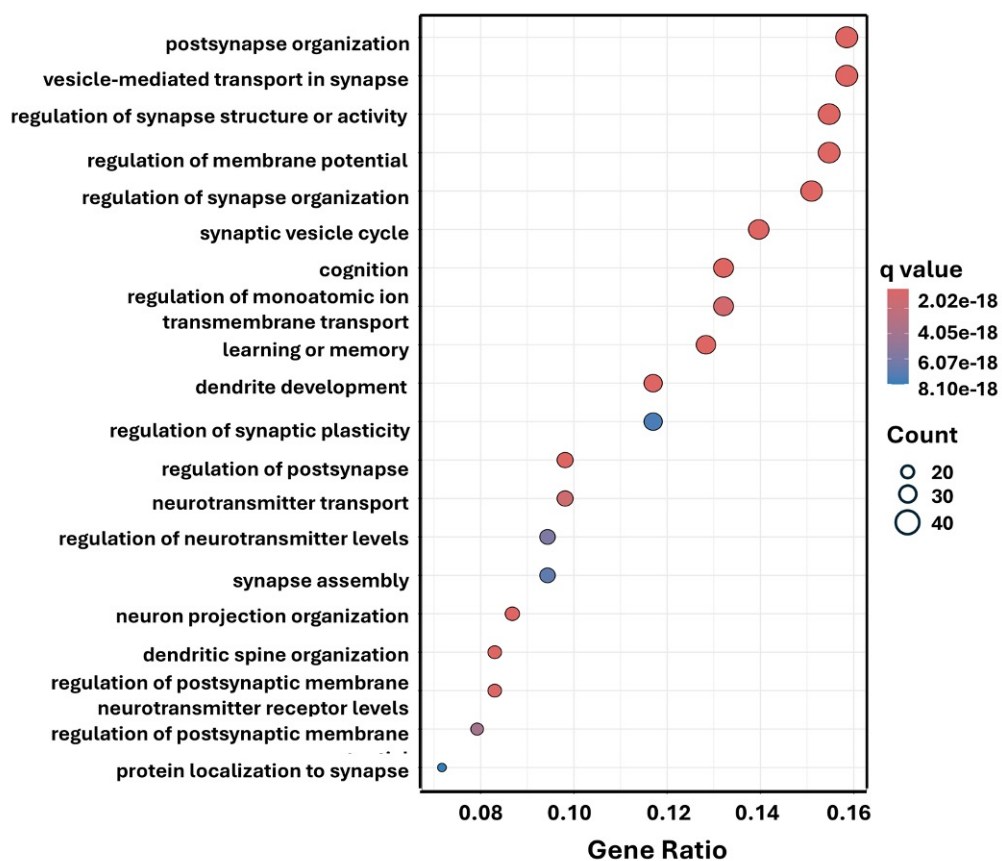


Cell-cell communications

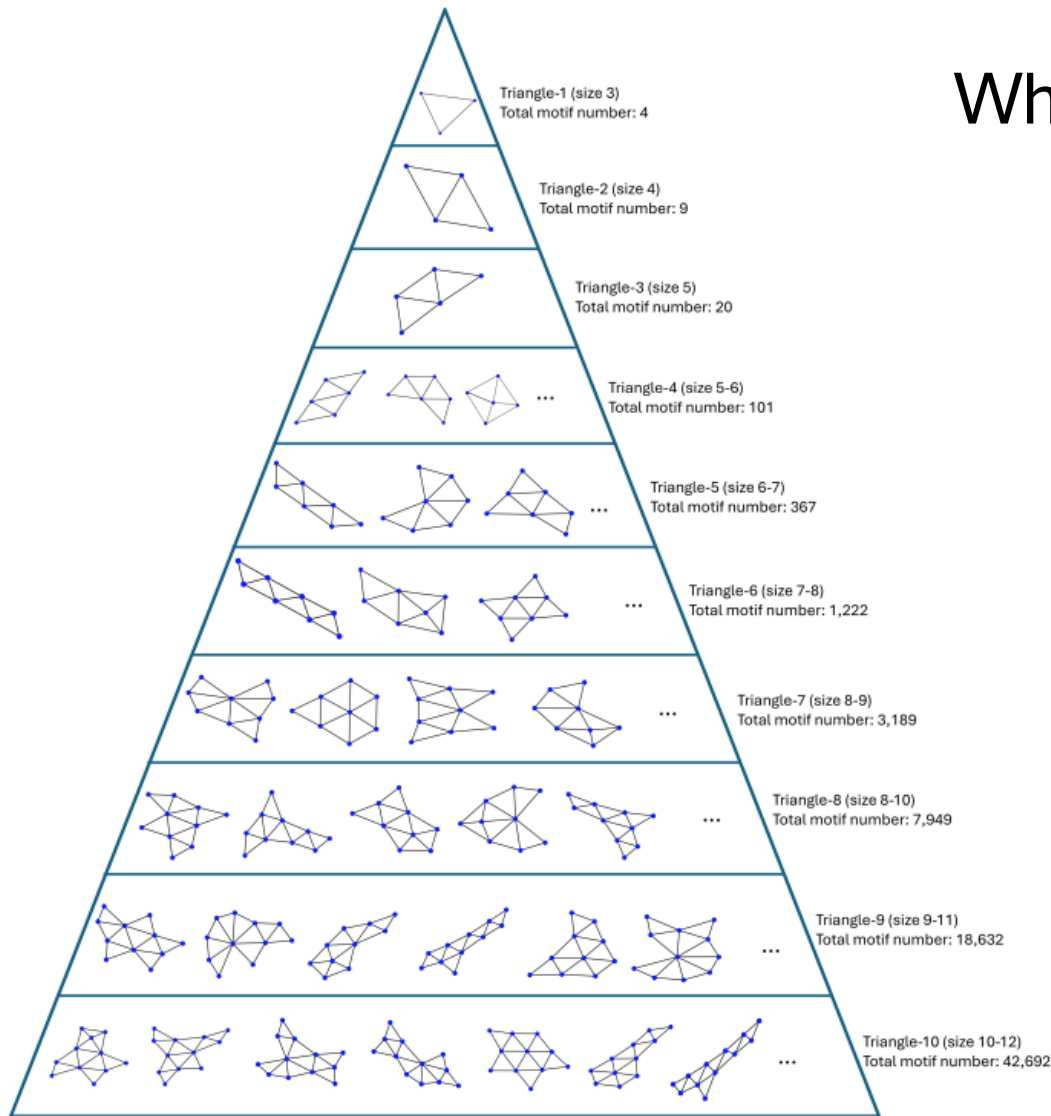


Expression patterns

Validated by GO and Pathway analysis

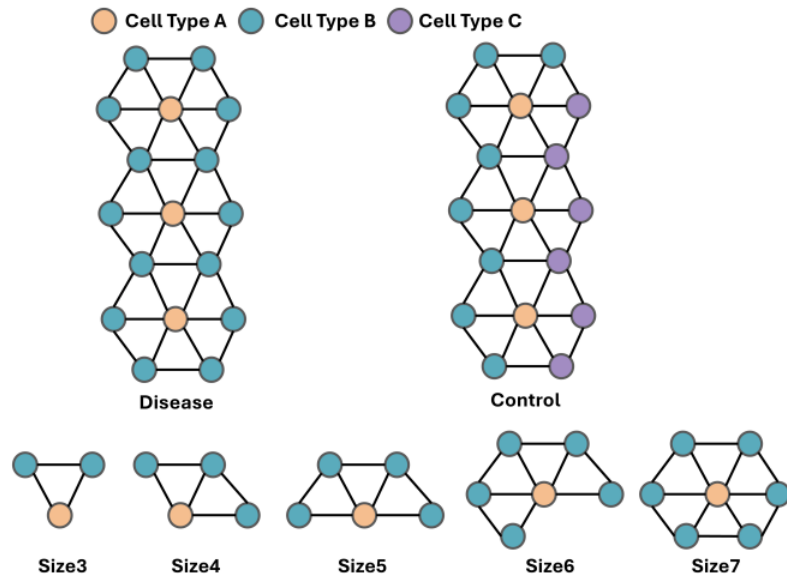


Which size is more appropriate



- Topology of CC motifs composed of triangles from **Triangle-1 to Triangle-10**. The total motif number denotes the number of unique CC motifs within two cell types, such as 'CTXEx' and 'Microglia' as node types in the applied Alzheimer's disease research.
- Label types: 2

A simulated toy example

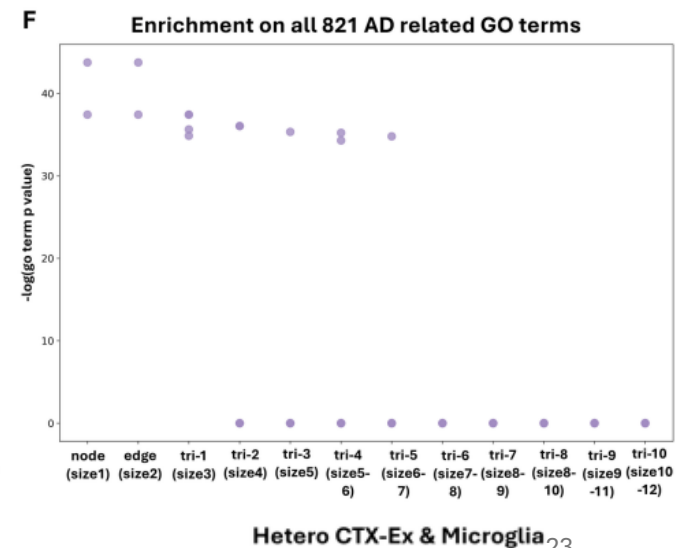
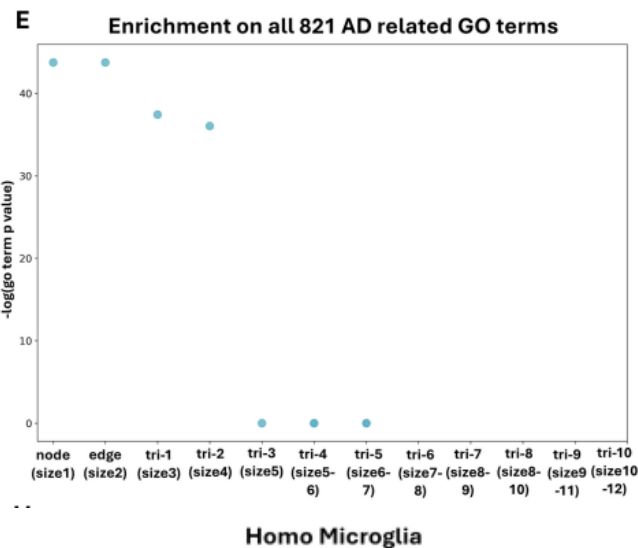
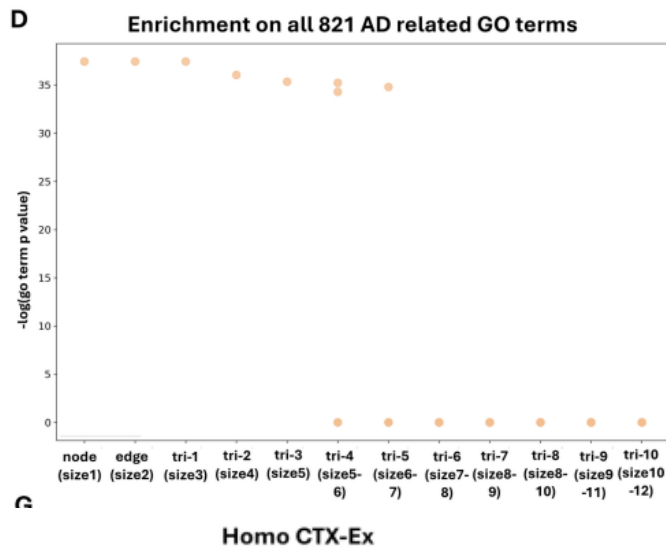


- Size 6 is selected

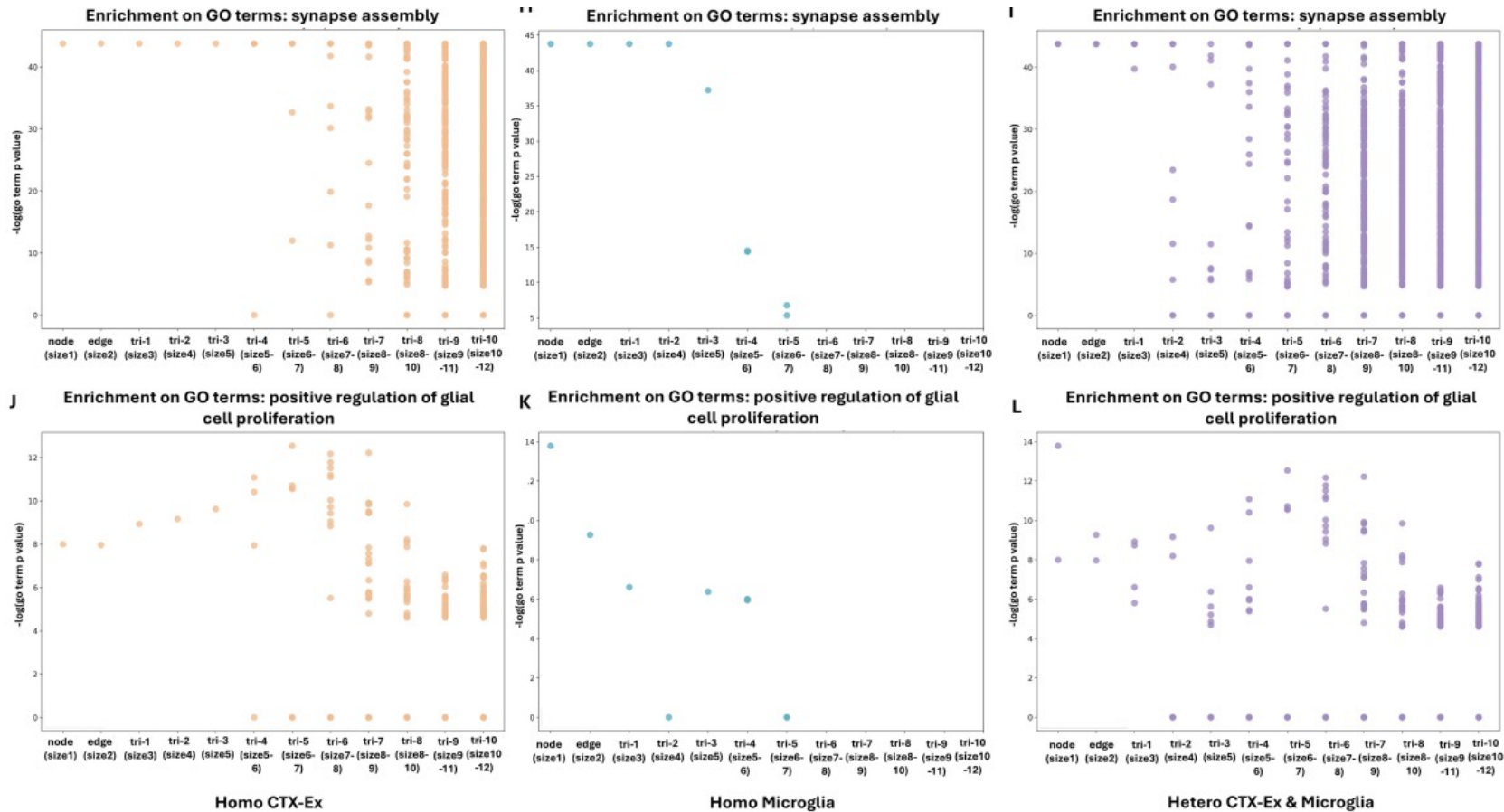
size	Disease	Control	Diff	Ratio (Dsease+1/Control+1)
Size-3	18	9	9	1.90
Size-4	18	6	12	2.71
Size-5	18	3	15	4.75
Size-6	18	0	18	19.00
Size-7	3	0	3	4.00

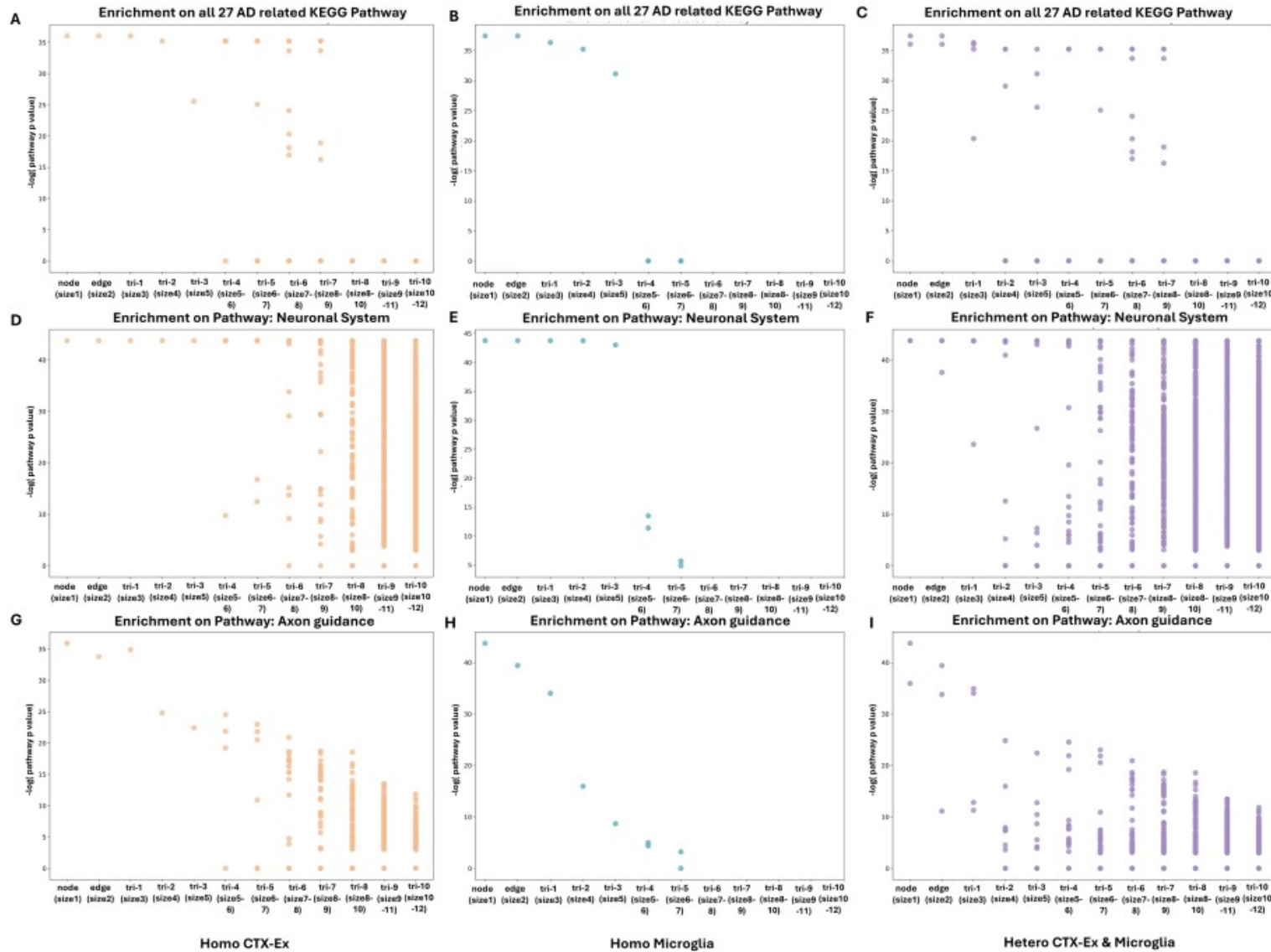
Evaluate which size is more appropriate

- No appropriate statistical-rigor approaches across different sizes
- Surrogating enrichment P-values of GO terms
- Meta-results



Diverse size for different purposes





- Similar in pathway analysis

Comparison of available computational methods

	Top-down	Bottom-up		
Approach	Clustering	Enumeration	Permutation	TrimNN
Interpretability	Low	High		
Generalizability	Low	High		
Designated sizes	No	Yes		
Size 1-3 Accuracy	NA	Ground truth	Low	High
Size 1-3 Speed	Very Fast	Fast	Fast	Fast
Size 4 and up Accuracy	NA	Not feasible	Low	High
Size 4 and up Speed	Very Fast	Not feasible	Fast	Fast

Taking home message

- CC motifs are data-driven and related to FTU
- CC motifs provide an expressive representation of the cellular neighborhood
- Interpretability and generalizability
- TrimNN et al. provide diverse deep learning approaches making counting CC motifs computationally feasible
- Potential usage in translational medicine?

Yu, Y., Wang, S., Li, J. *et al.* TrimNN: characterizing cellular community motifs for studying multicellular topological organization in complex tissues. *Nat Commun* **16**, 7737 (2025).

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Organoid Study

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Epilepsy Study

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AD Study

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